

Janak Kabra

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EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

Vellore, Tamil Nadu

Aug. 2024 – May 2028

EXPERIENCE

Software Intern

Feb. 2026 – Jun. 2026

Infosys Springboard

Remote

- Owned feature engineering and model selection within a project team, building LSTM and multivariate-regression forecasting pipelines and driving adoption of the lowest out-of-sample RMSE specification.

AI Research Intern

Oct. 2025 – Dec. 2025

Mythus AI Studios

Remote

- Partnered with a cross-functional ML team to certify classification models for production, and cut pipeline runtime by ~20% with vectorized NumPy and multiprocessing.

Selected Builder (competitive cohort)

Swiggy Builders Club

Remote

- Selected into Swiggy's competitive builder program; built SnapOrder, a live multimodal AI ordering agent (Next.js, TypeScript, LLM-driven tool use) that turns a food photo into a placed order.

PROJECTS

CryptoVault: Crypto-Agile Secrets Vault | [GitHub](#) | [Live Demo](#) | Java 21, Spring Boot, MySQL, Redis

- Built a crypto-agile secrets vault implementing KMS-style patterns from scratch: HKDF-SHA256 envelope encryption, swappable AES-256-GCM / ChaCha20-Poly1305 ciphers, and versioned key rotation with background re-encryption so legacy records still decrypt.
- Hardened access with stateless JWT plus Redis-backed revocation, RBAC, IP/email brute-force rate limiting, and a from-scratch RFC 6238 TOTP MFA with hashed one-time recovery codes.
- Enforced a Flyway-owned schema and full audit trail, verified by 60+ JUnit and Testcontainers integration tests (real MySQL + Redis) in CI, and deployed live across Vercel, Render, Aiven, and Upstash.

Volatility Surface Engine | [GitHub](#) | [Live Demo](#) | Python, Numba JIT, FastAPI, Docker

- Built an options pricing, calibration, and risk engine from first principles, implementing a Heston stochastic-volatility Fourier solver and an arbitrage-free SVI implied-volatility surface.
- Resolved the "Little Heston Trap" instability at long maturities with a branch-cut-free complex-logarithm representation, achieving stable desk-style calibration in under 100ms.
- Exposed full Greeks and surface outputs for hedging workflows via a Dockerized FastAPI service with CI and a 122-test suite.

NSE Factor Screener | [GitHub](#) | [Live Demo](#) | FastAPI, PostgreSQL, React

- Cut multi-factor screening latency to under 500ms over NIFTY-500 (P/E, ROE, Debt/Equity) via targeted indexing and single-round-trip queries.
- Engineered a peer-relative sector ranking engine computing cross-sectional percentiles, replicating the factor-normalization step in standard quant-equity workflows.

raftkv: Linearizable Distributed KV Store | [GitHub](#) | [Live Demo](#) | Go

- Implemented Raft consensus from scratch (election, replication, snapshotting, linearizable reads) backing a fault-tolerant KV store; verified linearizable under seeded Jepsen-style fault injection, with read $\approx 143\mu\text{s}$ / write $\approx 1.2\text{ms}$ and the full suite passing under the Go race detector.

TECHNICAL SKILLS

Languages: Java, Python, SQL, Go, TypeScript/JavaScript, C++, C

Backend & Data: Spring Boot, Spring Security, REST API design, FastAPI, PostgreSQL, MySQL, Redis, Flyway, Docker, Linux, Git, CI/CD

Security: Applied cryptography (AES-GCM, ChaCha20, HKDF), JWT, RBAC, TOTP MFA, envelope encryption

Financial Concepts: Black-Scholes & Heston pricing, implied volatility, option Greeks, factor models, time-series forecasting

CS Foundations: Data Structures & Algorithms, Operating Systems, DBMS, Distributed Systems, Concurrency